

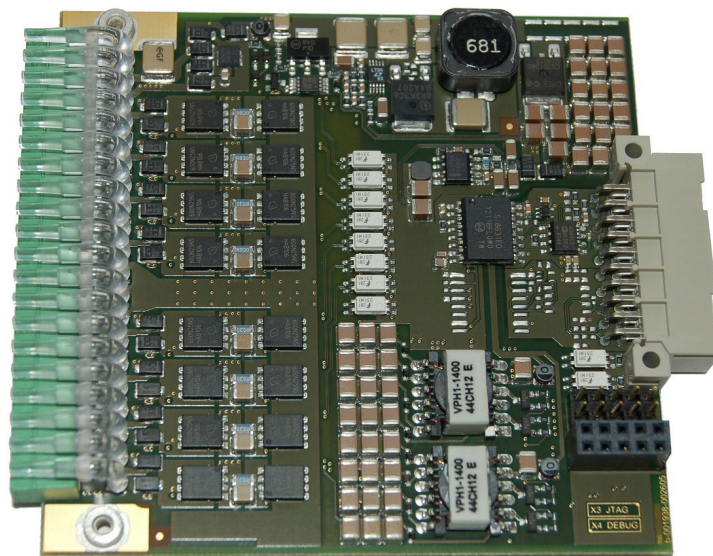


ionia™ i201 / i202 Rev3 Digital IO Module

Data Sheet

The i201/i202 Rev 3.0 is an intelligent digital I/O module for the ionia™ system. The i201 module has 16 digital input channels, whereas i202 module has 8 digital input channels and 8 combined input/output channels. The i202 module has output read back functionality for implementing safety relevant applications.

The i201/i202 Rev 3.0 is designed for harsh rolling stock environments and is compliant to EN50155.



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duagon Data Sheet Preamble

On having purchased products described in this data sheet, the customer acquires the right to use the products according to its specified purpose and in accordance with all operation, service and maintenance instructions. All other rights to the product, duagon's intangible assets rights in particular, belong solely to duagon and may not be deemed to have been assigned along with the sale of the products.

All product properties are fully described in the data sheet under express exclusion of any warranty for other properties. Of decisive relevance is the data sheet valid at the time of the order being placed. duagon provides a warranty that the product properties are retained during the period of warranty. Evidence that the properties of the product have been retained will be brought, always and exclusively, on duagon premises by means of a test construction pursuant to the type test.

The customer is obliged to inspect whether the products themselves are suitable for the application intended. In particular, that inspection must include the integration of the products into the intended system configuration and a check on whether the properties as per data sheet can be fulfilled once integrated into the system configuration as planned by the customer. Since the products are not certificated for operation with security applications, the customer must take appropriate measures to ensure that any malfunctions that may occur in a system configuration with other products will be absorbed by supplementary security measures.

The period of warranty for the products is 24 months and it begins on the date the products are shipped from the factory.

The warranty that duagon assumes for the products will, at duagon's discretion, be limited either to the repair of or the replacement of the products at the duagon factory. The warranty solely covers the products or parts thereof which, despite professional handling, have become defective or unusable and which arrive at the duagon factories for repair or replacement during the period of warranty. The extent of duagon's warranty is fully set out in this data sheet. duagon cannot be held liable for consequential damage caused by a defect or for indirect damage or for consequential damage of any kind. Therefore the customer bears all and any costs that occur due to production downtime, for example, or due to the installation or dismantling of products or due to their transportation to duagon and back.

duagon's liability and warranty do not obtain if evidence cannot be brought that the products were being operated according to its specified purpose and in accordance with all operation, service and maintenance instructions as issued by duagon.

These provisions form an integrated part of the product properties. duagon products cannot be acquired with other or more extensive degrees of warranty and liability on the part of duagon.

This data sheet is to be evaluated in accordance with **Swiss law**. The court of jurisdiction is the **seat of the vendor**. The applicability of the UN agreement as to international sales of goods (also known as "Viennese Purchasing Convention") is herewith expressly excluded.

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2						
3						

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1. General Safety Summary

Review the following safety precautions to avoid injury and prevent damage to this product or those connected to it. To avoid potential hazards, use this product only as specified.

Only qualified personnel should perform installation, maintenance and service procedures

To avoid fire or personal injury

Connect and disconnect properly. This data sheet contains all relevant information for connecting the device.

Power On only with all connections made. All connectors on the device must be connected (unused connections should be covered with a dummy connector).

Ground the product. Ground connection is located on the device.

Observe all terminal ratings. To avoid fire or shock hazard, observe all ratings and markings on the product. Consult this data sheet before making connections to the product.

Avoid exposed circuitry. Do not touch exposed connections and components when power is present.

Do not operate with suspected failures. If you suspect there is a damage to this product, have it inspected by qualified service personnel, or return it to duagon AG.

Safety Terms and Symbols

Terms in this data sheet. These terms may appear in this data sheet.



WARNING. Warning statements identify conditions or practices that could result in injury or loss of life.



CAUTION. Caution statements identify conditions or practices that could result in damage to this product or other property.

Service Safety Summary

Only qualified personnel should perform installation, maintenance and service procedures

Do not service with Power On. Dangerous voltages or currents may exist in this product. Disconnect power, remove battery (if applicable), and disconnect test leads before removing protective panels, soldering, or replacing components.

To avoid electric shock, do not touch exposed connections.

2. Introduction

General Features

The i201 and i202 are intelligent digital I/O modules for the ionia™ system. The i201 module has 16 digital input channels, whereas i202 module has 8 digital input channels and 8 combined input/output channels. The i202 module has output read back functionality for implementing safety relevant applications.

All high current inputs (HCI) provide a sampling mechanism to discharge parasitic capacitances. The HCI feature can be switched on and off by the application software. Additionally, selected inputs can be configured as frequency or counter inputs.

Every output is short-circuit protected and reports directly to the central processing unit. Moreover, selected output channels can be configured as PWM output channels.

The i201 and i202 modules are designed for harsh rolling stock environments and is fully to EN 50155.

2.1 How to Get Started

This document describes the hardware and soft-IP functionality of the i201 and i202 modules.

These modules are part of the modular ionia™ system, they are not intended to be used as standalone components.

ionia™ system-level

The ionia™ system is configurable online at <http://ionia.duagon.com/>

For information about programming the ionia™ system, please refer to:

- ionia™ Get Started Documentation (d-007724-nnnnnn).
- ionia™ IEC61131 API documentation (d-006935-nnnnnn).

i201 and i202 module-level

The i201 and i202 come with several order options, these are described in the section "Ordering Information", pag. 19. Not all options are available for the rev 3.0 boards.

2.2 Online Support

For additional information, please visit our home page www.duagon.com.

There you will find:

- Up to date documents
- Frequently asked questions
- Description of new product versions

3. Software considerations

3.1 Programming

The module is programmable by the central processing unit using a comfortable programming environment that is based on **IEC 61131 LD, FBD, ST, IL or SFC**.

Please refer to:

- ionia™ Get Started Documentation (d-007724-nnnnnn), and
- ionia™ IEC61131 API documentation (d-006935-nnnnnn).

3.2 Firmware update

Firmware update is handled by the central processing unit (*see i101/i102/i103 for further information*).

Thanks to a flash memory that stores a backup firmware image the i201 and i202 modules can be made power-fail-safe during a firmware upgrade process.

3.3 Backplane communication

The ionia™ system uses a duagon star slave (DSS) backplane communication.

Each module is connected with the central processing unit over a dedicated bidirectional link. The link is used to exchange all of the information with central unit. It uses a dedicated secure protocol. It is supervised with dedicated watchdog.

Central processing unit can scan the status of the modules in user-defined intervals (even down to cycle times of 1ms). The central processing unit can access to diagnostic information of the module as well (*see i101/i102/i103 for further information about these possibilities*).

4. Architecture

4.1 i201, i202 Hardware Structure

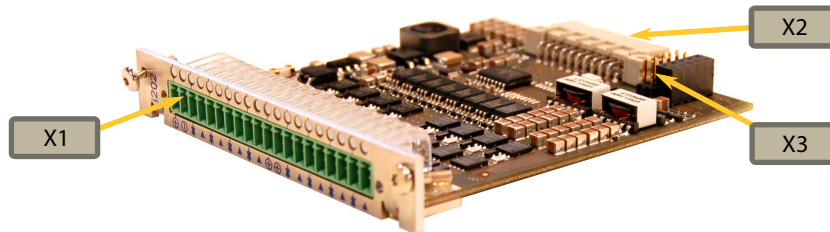


Figure 1: Hardware structure of the i202 module
(the i201 module has a similar outline).

Figure 1 shows the i202 hardware structure as a reference. The modules provide the following interfaces:

- X1: Front panel connector.
- X2: Backplane connector, and
- X3: Programming connector.

4.2 Basic Function Blocks

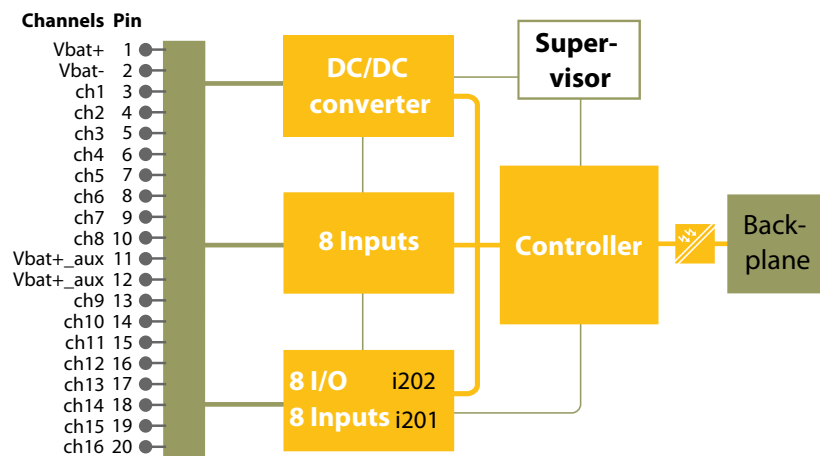


Figure 2: Schematic of the i201/202 architecture

The **Controller** contains the local microprocessor and all relevant logic circuitry, like memory controller, IO controller, etc. The controller is implemented inside a programmable logic device. This ensures the versatile change of, for example, controller specifications, bug fixes after hardware production, and future portability to new silicon technologies.

The **digital inputs** are referenced to the supply voltage (V_{BAT}). Nominal supply voltage is 24V¹. The digital inputs can be configured to operate in different modes, as summarized in section "Operating modes for digital inputs and outputs" on pag.8.

The **digital outputs** are high side FET switches referenced to the supply voltage (V_{BAT}). The digital outputs can be configured to operate in the different modes summarized in section "Operating modes for digital inputs and outputs" on pag.8.

1) Other nominal voltage ranges like 48V, 72V, 96V or 110V will be supported by future revision (Rev. 5)

Power is drawn directly from the battery. In the complete system, every single i201 or i202 can be supplied with other supply voltages circuit. The needed internal logic voltages are generated internally out of the battery voltage.

Operating modes for digital inputs and outputs

The following two tables summarizes the different operating modes available on the digital input and output channels.

Digital input channel				REV 3
Mode	See pag. ...	HC I	Description	
Normal	9	O	Normal input operating mode	Supported
Frequency – time window mode		P	Used to count the number of impulses within a specified time-window	Not supported in Rev 3.0, only available in Rev 5
Frequency – edge detect mode		P	Used to measure the time between two consecutive impulses	Not supported in Rev 3.0, only available in Rev 5

Digital output channel				REV 3
Mode	See pag. ...	HC I	Description	
Normal	11	P	Normal output operating mode	Supported
Pulse Width Modulation (PWM)		O	Output is configured to drive a PWM signal	Not supported in Rev 3.0, only available in Rev 5

These operation modes can be complemented by additionally enabling or disabling the HCI functionality. The column “HCI” in the table above shows if the operation mode can be combined with HCI functionality, O: can Optionally be combined with HCI functionality; P: is Prohibited.

The HCI functionality is related to the digital inputs and therefore it is described in the digital input part, section “High Current Input (HCI) functionality” pag. 10.

4.3 Digital inputs – i201 and i202 modules

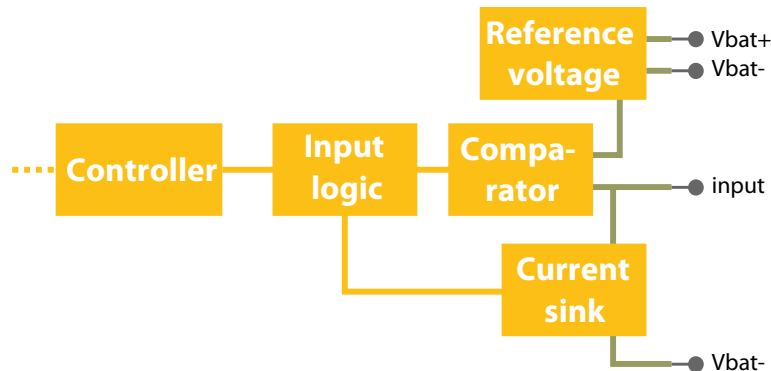


Figure 3: Input diagram

Figure 3: Input diagram, depicts the block diagram of the digital input circuitry.

Digital **inputs** receive a digital signal applied to the pin. If the channel is shared with a digital output (in the case of the i202 module), it can be used as read-back. The read-back functionality is useful, for example, to check that the corresponding output effectively is in the desired state. The inputs are referenced to Vbat+ and Vbat-.

A **comparator** detects the input voltage levels and adapts the levels for the input logic block.

The **controller** samples the input logic and processes the input signal according to the processing mode currently selected (i.e., normal operation or frequency measurement). An internal glitch-filter is used to filter out short glitches that are present on the input signal. The central processing unit of the ionia™ system accesses the module's controller via the backplane and feeds it with appropriate commands.

The **current sink** block is used in conjunction with the HCI functionality. When the HCI functionality is enabled, short current pulses are drawn from the input. This functionality is useful for discharging parasitic capacitances, etc.

The digital input operation modes and the HCI functionality are described in the next sections.

4.3.1 Normal operation mode

In this operation mode, the status of the digital input pin is read by the controller and made available to the CPU. All input channels (see also section "Front Panel" pag.12 for the information about the pin to channel mapping) can operate in this mode. The CPU accesses the i202 module over the backplane and – through appropriate commands – reads out the status of the inputs.

If the input is higher than V_{INH} then the input is regarded as a logic '1'. Otherwise, if the input is lower than V_{INL} then the input is regarded as a logic '0'.

4.3.2 High Current Input (HCI) functionality

When the HCI functionality for one input or input only channel is enabled by the application on the CPU, the current sink block of the corresponding channel is enabled. The functionality is described in the following.

Input only channels

When a logic '1' is applied to the pin of the input channel, the current sink block starts to sink current. To reduce power consumption high current is sunk only in short periods of time.

When a logic '0' is applied to the pin of the input channel, the current sink block is disabled.

In Figure 4, the parameters t_{IN_H} , t_{IN_L} , I_{INH} and $I_{INH, idling}$ are described. For the electrical specification and the exact values see section Digital input channels on pag.16.

This operation mode can be enabled or disabled (default) by the application running on the CPU (see also the allowed combinations reported in Operating modes for digital inputs and outputs on pag.8).

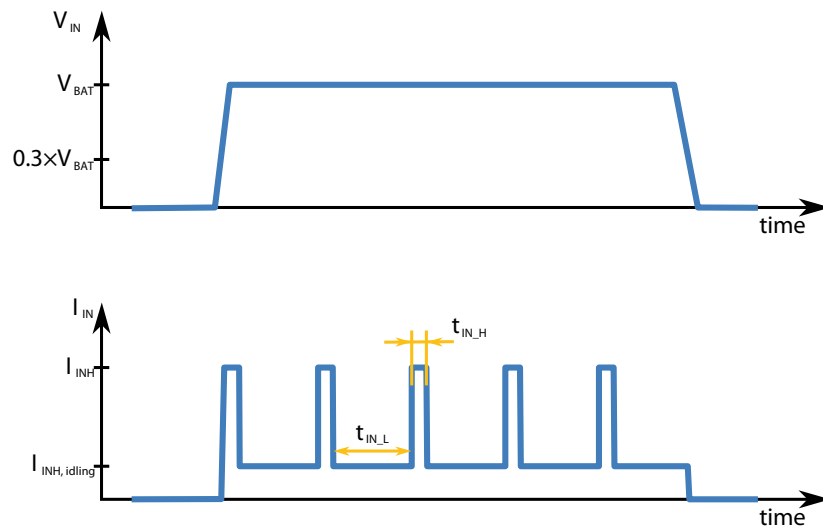


Figure 4: HCI operation

4.4 Digital outputs – i202 modules

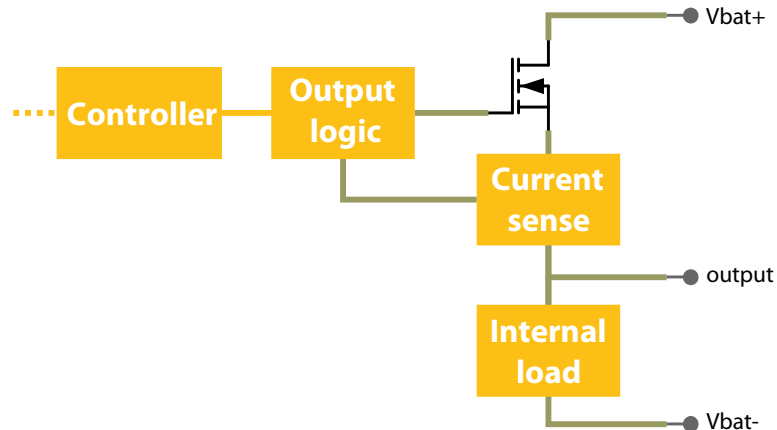


Figure 5: Output diagram

The output channels are referenced to Vbat+ and Vbat-. Each output channel has an individual **current sense** circuitry, which switches the output off when detecting an over-current.

The **current sense** circuit latches the over-current state and, to return into normal operation, the output channel needs to be re-enabled by the CPU application.

The **digital output** channel FET is either enabled – and hence the output pin is at Vbat+. Otherwise, the output channel FET is open and the output pin is tight to Vbat- through an internal load (R_{INT_LOAD} , see also section “Digital output channels” pag.16).



CAUTION. In order to give maximum flexibility in implementation, outputs are not fused. It is advised to apply external fuses on Vbat+ and/or outputs in case of unlikely failure of over-current/switching circuit.

4.4.1 Normal output mode

In this operation mode, the pin of the digital output channel is either strongly tight to Vbat+, or, it is weakly tight to Vbat-. When the CPU application drives a logic '1', the output channel FET is enabled – and hence the output pin is at Vbat+. Otherwise, when a logic '0', the output channel FET is open and the output pin is tight to Vbat- through an internal load (R_{INT_LOAD}).

5. Physical Interfaces

5.1 Front Panel

The front panels of the i201 and i202 are shown in Figure 6.

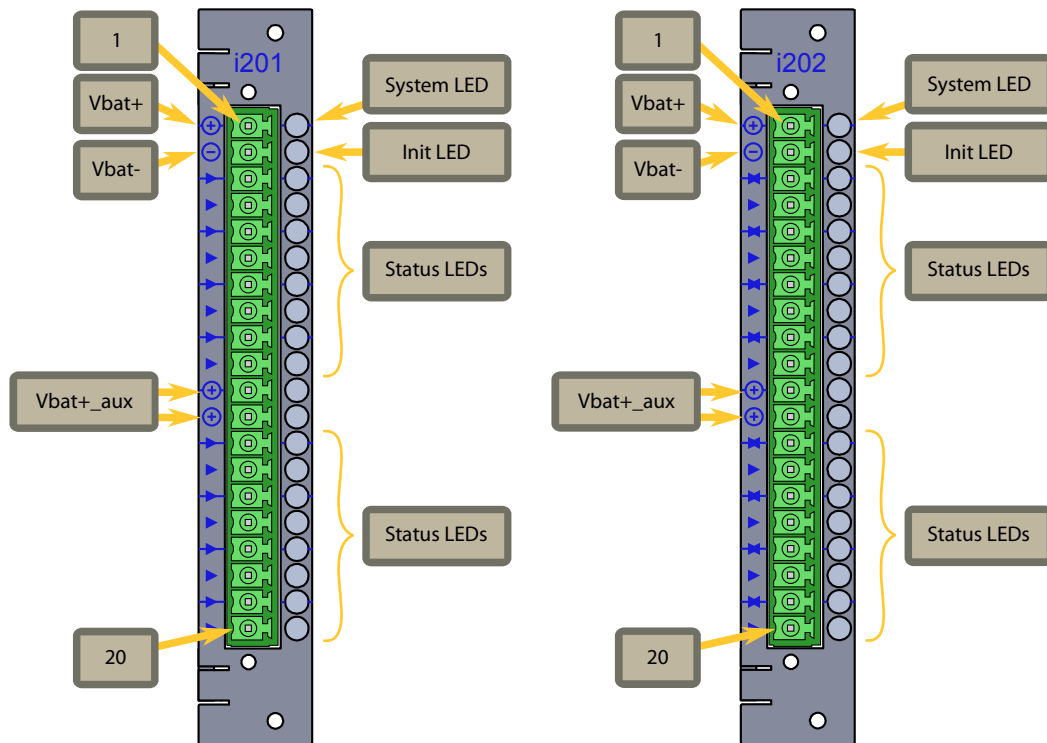


Figure 6: Front panel of the i201 (left) and i202 (right) modules.

5.1.1 Front panel connector (X1) – i201 module

Pin #	Signal Name	Description
1	Vbat+	Battery input voltage positive pole
2	Vbat-	Battery input voltage negative pole
3, 5, 7, 9	ch1, ch3, ch5, ch7,	Input channels 1, 3, 5, 7
13, 15, 17, 19	ch9, ch11, ch13, ch15	Input channels 9, 11, 13, 15
4, 6, 8, 10,	ch2, ch4, ch6, ch8,	Input only channels 2, 4, 6, 8 – Freq. meas. group 1
14, 16, 18, 20	ch10, ch12, ch14, ch16	Input only channels 10, 12, 14, 16 – Freq. meas. group 2
11, 12	Vbat+_aux	Auxiliary battery input voltage positive pole, may be left open

5.1.2 Front panel connector (X1) – i202 module

Pin #	Signal Name	Description
1	Vbat+	Battery input voltage positive pole
2	Vbat-	Battery input voltage negative pole
3, 5, 7, 9	ch1, ch3, ch5, ch7,	Input / Output channels 1, 3, 5, 7 – PWM group 1
13, 15, 17, 19	ch9, ch11, ch13, ch15	Input / Output channels 9, 11, 13, 15 – PWM group 2
4, 6, 8, 10,	ch2, ch4, ch6, ch8,	Input only channels 2, 4, 6, 8 – Freq. meas. group 1
14, 16, 18, 20	ch10, ch12, ch14, ch16	Input only channels 10, 12, 14, 16 – Freq. meas. group 2
11, 12	Vbat+_aux	Auxiliary battery input voltage positive pole, may be left open if current requirements are met

5.1.3 Cabling

The input/output connector on the front panel (**X1**) is mating with Phoenix Contact FMC 1,5/20-STF-3,5 (PN: 1966279) or compatible.

A wire with cross from 0.2mm² to 1.5mm² can be used.

5.2 LED Status indication

LED	Colour	Status
System	green	ON – Power OK Blinking – backplane communication
	red	Error detected (over-current or I ² C/SPI problem)
	off	No power
Init	yellow	Initialization, PLL lock
	off	No power / normal operation
Status	yellow	Input active (default)
	off	Input inactive / no power (default)

user-defined¹⁾

Note:

- 1) Each status LED can be individually controlled by the user.

5.3 Backplane

5.3.1 Backplane connector (X2)



CAUTION. The backplane interface is intended for internal backplane connection inside ionia™ system. Any other use may lead to malfunction or damage of ionia™ system.

Pin out of the connector is provided for reference only.

Pin #	Signal Name	Description
a5, b5, a6, b6	NC	Not connected
a4, b4	3.3V_A	Backplane auxiliary power input channel A
a7, b7	3.3V_B	Backplane auxiliary power input channel B
a1, a2, b3, b8, a9, a10	GND	Backplane ground
b1	data_Ap	Backplane communication channel A, positive
b2	data_An	Backplane communication channel A, negative
b9	data_Bp	Backplane communication channel B, positive
b10	data_Bn	Backplane communication channel B, negative
a3	Burn-in	Enable burn-in test mode
a8	FuncTest	Enable functional test mode

5.4 JTAG Interface (X3)

The JTAG interface provides an interface for debugging and firmware upgrades.



CAUTION. The JTAG interface is intended for service purposes only. They shall not be used for normal operation. The JTAG interface is available only as internal connector. There is no possibility and no need to access the JTAG from outside the ionia™ housing. All firmware updates can be made via the backplane connection.

Pin #	Signal Name	Description
1	TCK	JTAG signal line
2	GND	Local power reference
3	TDO	JTAG signal line
4	+3.3V	Local power line
5	TMS	JTAG signal line
6	+3.3V	Local power line
7	NC	Not connected / Do not connect
8	NC	Not connected / Do not connect
9	TDI	JTAG signal line
10	GND	Local power reference

6. Electrical specification

typical values for $T_A = 25^\circ\text{C}$, $V_{BAT}=24\text{V}$ (unless otherwise specified).

6.1 Power supply

Symbol	Parameter /Conditions	Min.	Nom. / Typ.	Max.	Unit	EN50155:2007
V_{BAT}	Supply voltage (for Rev 3.0)	16.8	24 ¹⁾	33.3	V	5.1.1.1 Variations of voltage supply
$V_{BAT,non\ oper}$	Non operating supply voltage	-33.3		33.3	V	7.2.6 Polarity reversal
$I_{BAT, fuse}$	Absolute maximum input current of logic (fuse)		0.375 ²⁾		A	
I_{inrush}	Inrush current (peak)	2 ³⁾		20 ³⁾	A	7.2.7 Inrush currents
I_{pin_max}	Current over a single Vbat+ pin			6 ⁴⁾	A	
V_{UVLO}	Under voltage lock out			10	V	
P_{idle}	No-load input power	$V_{BAT}=24\text{V}$, i201	0.7		W	5.1.1.1 Variations of voltage supply
		$V_{BAT}=24\text{V}$, i202	1.2		W	
t_{S2}	Hold-up time after input interruption	10	15		ms	5.1.1.2 Interruptions of voltage supply, Class S2

Notes:

- 1) Further nominal voltages listed in EN50155 are supported in Rev 5.
- 2) Power supply must be able to deliver enough current to break the fuse in case of short circuit.
- 3) Values obtained during type testing. Do not use them as absolute extremes. They depend on the applied V_{BAT} .
- 4) There are three Vbat+ connections provided. Appropriate number of them shall be used to provide sufficient current to the module.

6.2 Digital output channels

Symbol	Parameter /Conditions		Min.	Nom. / Typ.	Max.	Unit
$R_{OUT,on}$	Output switch on resistance			0.2	0.5	Ω
I_{OUT}	Output current on single pin	continuous	0 ¹⁾		1	A
		Surge peak, dropping over 100ms to continuous value			6	A
$P_{OUT,tot}$	Total continuous output power of all output channels				500	W
t_{rise}	"0" to "1" transition time	$I_{OUT}=1A$		0.6		μs
$t_{fall}^{2)}$	"1" to "0" transition time	$I_{OUT}=1A$		0.2		μs
t_{ON}	Minimum ON time		10			μs
t_{OFF}	Minimum OFF time		10			μs
$R_{INT,LOAD}$	Internal resistive load, no HCI			334		k Ω

Notes:

- 1) No negative current is allowed.
- 2) Fall time depends on attached load. The use read-back with HCI enabled helps to reduce fall time with light loads.

6.3 Digital input channels

Symbol	Parameter /Conditions		Min.	Nom. / Typ.	Max.	Unit
V_{INH}	High input voltage		$0.3 \times V_{BAT}$		V_{BAT}	V
V_{INL}	Low input voltage		0		$0.17 \times V_{BAT}$	V
I_{INH}	Sink current for high input voltage		8	10	15	mA
$I_{INH, idling}$	Sink current for high input voltage, idling	$V_{BAT} \pm 24V$	0.014	0.024	0.2	mA
$t_{IN,H}$	Current sink duration			33		μs
$t_{IN,L}$	Current sink duration, idling			967		μs

7. Environmental Specification

Issue	min.	typ.	max.	Unit	EN50155:2007
Operational temperature range, "internal cubicle temperature",	-40	+25	+70	°C	4.1.2 Class T2
Air temperature surrounding the PCB	-40	+25	+70	°C	4.1.2 For information, only.
Altitude above sea level			1800	m	4.1.1 For derating of maximum temperature with respect to altitude levels in excess please call duagon.
Relative humidity		<75	95	%	Typical value for yearly average, max value for 30 consecutive days per year. Short term moisture condensation without malfunction (DIN EN 60068-2-30:2006-05)
Vibration			2 50	mm m/s ²	12.2.11 for 5-25 Hz for 25- 150 Hz
Shock			50	m/s ²	Duration 50 ms
RFI susceptibility from 0.15 to 2000 MHz.	20			V/m	

8. Mechanical Data

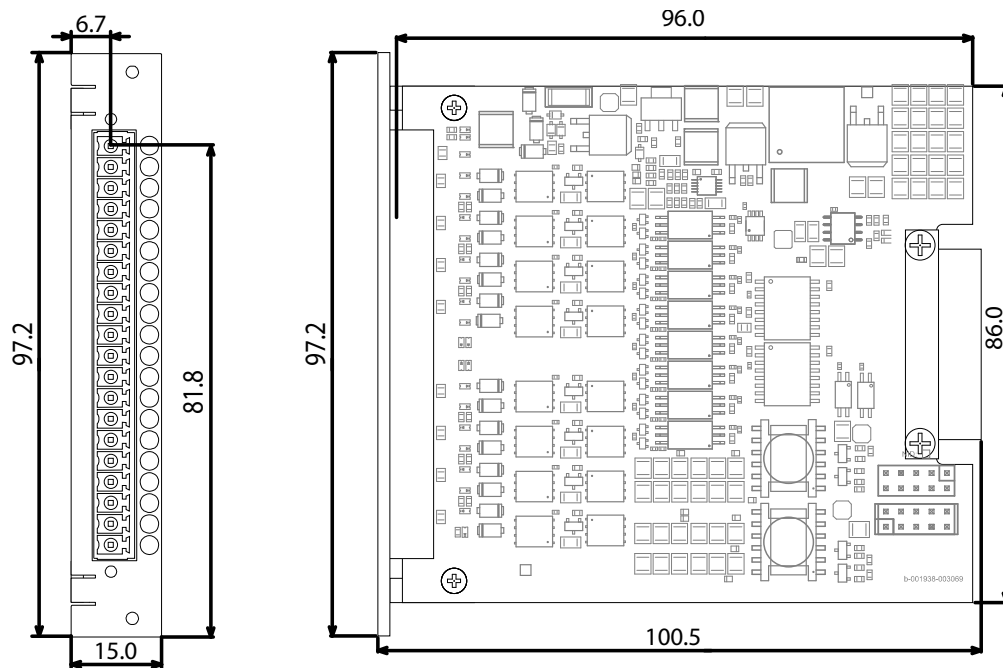


Figure 7: Dimensions of the i201 and i202

		Unit
Weight (i201)	88 ^{1) 2)}	g
Weight (i202)	99 ^{1) 2)}	g
PCB dimensions	96 × 86	mm

Notes:

- 1) Weight includes metal front panel.
- 2) Weight does not includes conformal coating.

9. Ordering Information

In the blow table the different order options can be found. Please consider to use the online configurator (ionia.duagon.com) for a simple configuration.

i	202	-F	-P
Platform: Platform	ionia™		
Product Type: 16 digital inputs	201		
8 digital inputs and 8 digital inputs/outputs	202		
Frequency measurement: no		-N	
on 8 channels		-F	
PWM: (i202 only) no			-N
on 8 channels			-P

Option F and P are not available in Rev 3. Take Rev 5 to get this options.

10. Normative Reference

The i201 and i202 complies to the standards **EN50155**, and **EN50121-3-2** in general. For exceptions, not applicable sub-clauses etc. please refer to the following paragraphs and, in more detail, to the corresponding type test reports "i201 and i202 Type Test Top – Report" with the document number d-006906-nnnnnn:

Immunity

EN50121-3-2:2006 7.1 Battery referenced ports: Conducted radio frequency *EN61000-4-6*, 10Vrms (carrier voltage), 150kHz – 80MHz, 1kHz, 80% AM, Source impedance 150Ω

EN50121-3-2:2006 7.2 Battery referenced ports: Fast transients bursts according to *EN61000-4-4*, ±2kV 5/50ns tr/th, 5kHz repetition frequency.

EN50121-3-2:2006 7.3 Battery referenced ports: Surges according to *EN61000-4-5*, waveform 1.2/50μs, wire vs grounding ±2kV 42Ω 0.5uF, wire vs wire ±1kV 42Ω 0.5uF

EN50121-3-2:2006 8.1 Signal I/O: Conducted radio frequency *EN61000-4-6* 10Vrms (carrier voltage) 150kHz – 80MHz, 1kHz, 80% AM, Source impedance 150Ω

EN50121-3-2:2006 8.2 Signal I/O: Fast transients bursts according to *EN61000-4-4*, 2kV ±5/50ns tr/th, 5kHz repetition frequency

EN50121-3-2:2006 9.1 Enclosure port: Radio frequency *EN61000-4-3*, 20V/m (rms carrier voltage) 80MHz – 1GHz, 1kHz 80% AM.

EN50121-3-2:2006 9.2 Enclosure port: Radio frequency *EN61000-4-3*, 20V/m (rms carrier voltage) 800MHz – 1GHz, 1kHz 80% AM; 10V/m (rms carrier voltage) 1.4GHz – 2.1GHz, 1kHz 80% AM; 5V/m (rms carrier voltage) 2.1GHz – 2.5GHz, 1kHz 80% AM;

EN50121-3-2:2006 9.3 Enclosure port: Electrostatic discharge *EN61000-4-2*, 6kV contact discharge, 8kV air discharge.

Insulation

EN50155:2007 12.2.9: Insulation measurement test / Voltage withstand test. 1000Vrms, 1MΩ for 1 minute. During routine testing, the testing is optionally changed to sinusoidal 1500Vrms rms, 1MΩ for 10sec. Each individual device is tested prior to shipment (routine testing).

Emission

EN50121-3-2:2006 4.1 and 5.1 Battery referenced ports, process measurement and control ports: *EN55011*, 150kHz – 500 kHz 99dBμV/m quasipeak; 500kHz – 30MHz 93 dBμV/m quasipeak.

EN50121-3-2:2006 6.1 Enclosure port: *EN55011*, 30MHz – 230 MHz 40dBμV/m quasipeak measured at 10m; 230MHz – 1GHz 47 dBμV/m quasipeak measured at 10m distance.

Useful Life

EN50155:2007 6.2: There are components with a limited useful life in the i201 and i202:

Reprogrammable components (Flash Memory). These are specified from the manufacturer with 20 years data retention at 125° Celsius, which fits clause 6.2. The actual "useful life" can be extended by reprogramming these devices.

Remark: There are no electrolytic capacitors, which typically introduce a limitation to useful life.

Polarity Reversal

EN50155:2007 7.2.6: Polarity reversal. The i201 and i202 are protected against polarity reversal with the restriction, that current limitation for the output pins is done on the outside of the module. This is typically the case by the normal load, which is operated with inverted voltage in this case. In order to keep this feature, do not use loads, which enter short-circuit when powered with the wrong polarity (Example: "free wheeling diodes" over relays coils).

Components

EN50155:2007 8.1.5 and 8.1.7: There may be specialized components/ single source components included in the product. Contact duagon for more information about repair and long term shipment procedures.

Shock and Vibration

EN61373:1999: There is no specific definition of the mounting direction in the vehicle, i.e. the highest requirement is to be applied to all directions of the i201 and i202. The i201 and i202 applies to mounting location "Category 1, Class B".

Degree of Protection

EN60529:2000: The degree of protection is IP 30.

Fire and Smoke

EN45545-2:2010: The i201 and i202 falls in the category of "not-listed" products. There are no special requirements for products with a mass lower than 100g (interior location) and 400g (exterior location).

Manufacturing

The manufacturing of the PCB assembly is done according to **IPC-A-610D:2005 level 2**.

The product complies to the European Union directive **EC/2002/95 (RoHS compliance)**.

REACH

Concerning the European Union directive **EC/1907/2006 (REACH compliance)**, duagon does not need to register any substance. duagon's products do not use quantities of more than 1 ton of a certain substance and the substances are not released under normal conditions of use. For a registration, both criteria would have to be fulfilled.

Humidity

EN60068-2-30:2006: Provides a composite test procedure, primarily intended for component type specimens, to determine, in an accelerated manner, the resistance of specimens to the deteriorative effects of high temperature/humidity and cold conditions.

11. Links to other duagon Documents

In general, most of the documents are located on www.duagon.com, and may be downloaded from there in the most up to date version.

i201 and i202 Short Form Data Sheet, d-006282-nnnnnnn

The ionia™ i201 and i202 intelligent digital IO modules short form data sheet. The short form data sheet is an abstract of this document and covers only general description and specification.

i201 and i202 Type Test Top – Report, d-006906-nnnnnnn

This document covers the type testing performed with the i201 and i202 modules.

ionia™ Get Started Documentation, d-007724-nnnnnnn

Document describing first steps for set-up and configuration of the ionia™ system.

ionia™ IEC61131 API documentation, d-006935-nnnnnnn

Document describing the IEC61131-based programming environment for the ionia™ system.

Labeling and Packaging – Specification, d-000778-nnnnnnn

This document describes all product labels (e.g. serial number label) used in relation with customers. It describes furthermore how duagon packs the products for shipment.

Quality Plan for duagon Products – Specification, d-000796-nnnnnnn

This document is a specification about test procedures for series testing of duagon products. It is valid for all duagon products in general. For each specific product an applicable subset of the described tests is selected; according to the procedures specified here.

Life Cycle of duagon's Products – Technical Note, d-000526-nnnnnnn

Addresses some MTBF and Life Cycle considerations. The i201 and i202 is not included in the list of devices, yet: The i201 and i202 may be added after having collected sufficient statistical data from the field. In a very general way, the "Life Cycle"- document may be used for planning the life time repair stock in order to ensure long support times. If further informations are needed on this topic please ask duagon for a MFTB calculation for the i201 and i202.

Material and Components for Wiring – Technical Note, d-000842-nnnnnnn

This document is intended for engineers to help them select the right connectors, etc.

12. Document History

d-010330-018360

- First preliminary release